

CCNA Mega Lab

Introduction

This lab was sourced from the Jeremy's IT lab YouTube channel and was created as a comprehensive resource to practice all the relevant configuration topics for the exam. The completion score is fairly inconsistent and doesn't always update after a step is completed.

The lab's topology is as follows:

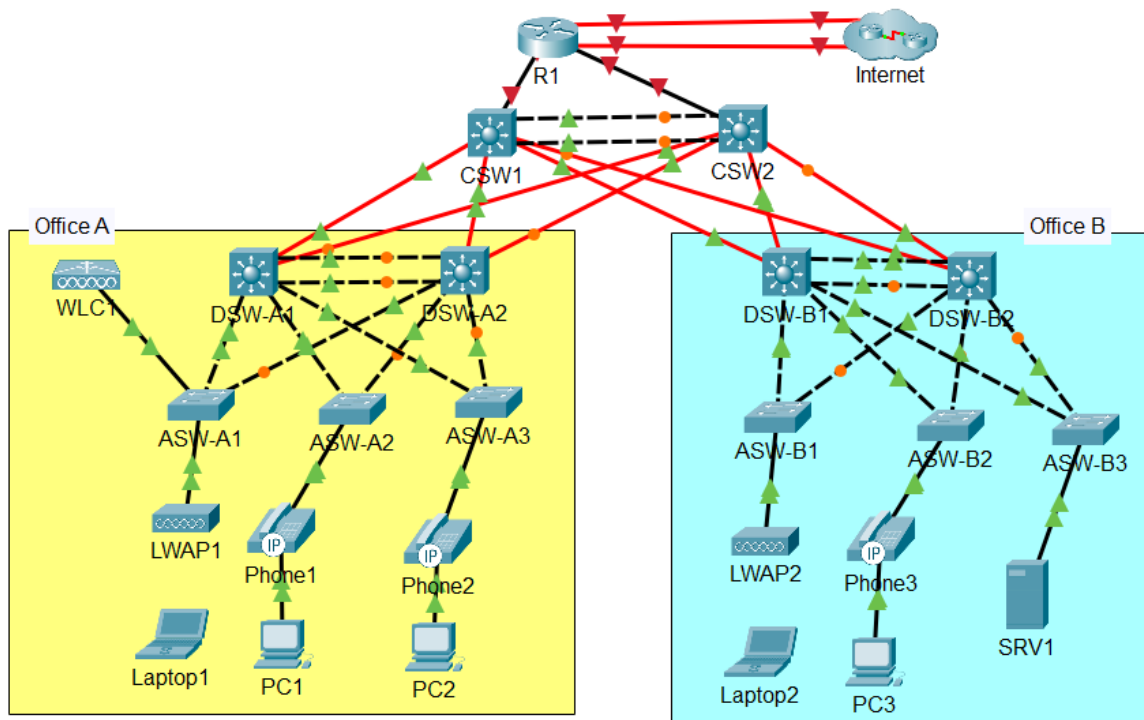


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1.0 Initial Setup

Initial Setup Includes the following steps

1. Configure the appropriate hostname on each router/switch.
2. Configure the enable secret `jeremysitlab` on each router/switch. Use type 9 hashing if available; otherwise, use type 5.
3. Configure the user account `cisco` with secret `ccna` on each router/switch. Use type 9 hashing if available; otherwise, use type 5.
4. Configure the console line to require login with a local user account. Set a 30-minute inactivity timeout. Enable synchronous logging.

1.1 Hostname

This subsection covers the following objective:

Configure the appropriate hostname on each router/switch.

On each router/switch:

Enable

Configure terminal

Hostname <HOSTNAME>

Do write

e.g. for router R1

enable

configure terminal

hostname R1

do write

1.1.1 Completion

Completion: 0%



Dock

Check Results

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#host
Router(config)#hostname R1
R1(config)#Do write
Building configuration...
[OK]
R1(config)#
```

Completion: 2%



Dock

Check Results

After repeating for all the other switches in the topology with the given hostnames

Completion: 22%



Dock

Check Results

1.2 Secret

This subsection covers the following objective:

Configure the enable secret jeremysitlab on each router/switch. Use type 9 hashing if available; otherwise, use type 5.

The core/distribution switches support type 9 hashing while the other devices only support type 5

1.2.1 Type 5 Hashing

Enable

Configure Terminal

Enable secret <SECRET>

Do write

e.g. for R1:

enable

configure terminal

enable secret jeremysitlab

do write

1.2.2 Type 9 Hashing

Enable

Configure terminal

Enable algorithm-type scrypt secret <SECRET>

e.g. CSW1

enable

configure terminal

enable algorithm-type scrypt secret jeremysitlab

1.3 User Account

This subsection covers the following objective:

Configure the user account cisco with secret ccna on each router/switch. Use type 9 hashing if available; otherwise, use type 5

Type 5 and type 9 hashing availability is the same as for the section that covers secret configuration

1.3.1 Type 5

Enable

Configure terminal

Username <USERNAME> secret <SECRET>

Do write

e.g. for R1:

enable

configure terminal

username cisco secret ccna

do write

```
R1>en
Password:
R1#
R1#en
R1#
R1#en
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#user
R1(config)#username cisco sec
R1(config)#username cisco secret ccna
R1(config)#do wr
Building configuration...
[OK]
```

```
R1(config)#do sh ru | i secret
enable secret 5 $1$mERr$LG1PCX0pnG7g1VqYOLG9a/
username cisco secret 5 $1$mERr$Bok4KdfVutXOJolNq009M/
```

The 5 after secret shows the hashing is type 5

1.3.2 Type 9

Enable

Configure terminal

Username <USERNAME> algorithm-type scrypt secret <SECRET>

Do write

e.g. for CSW1

enable

configure terminal

username cisco algorithm-type scrypt secret ccna

do write

```
CSW1>en
Password:
CSW1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
CSW1(config)#us
CSW1(config)#username cisco al
CSW1(config)#username cisco algorithm-type s
CSW1(config)#username cisco algorithm-type scrypt s
CSW1(config)#username cisco algorithm-type scrypt secret ccna
CSW1(config)#do wr
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
CSW1(config)#do sh run | i secret
enable secret 9 $9$J19FIAftPZf7c3$13koe1VgzliJg5KQU.JgBhXj4RzTxEUDznF2s1b.Hn6
username cisco secret 9 $9$J19FIAftPZf7c3$Mux3wTz1Kth/NxeyEk.E8.QBckaZchVfllsGnx08/0A
```

The 9 after secret shows the hashing is type 9

1.4 Timeout and Synchronous Logging

This subsection covers the following objective:

Configure the console line to require login with a local user account. Set a 30-minute inactivity timeout. Enable synchronous logging.

Enable

jeremysitlab

Configure terminal

Line console 0

Login local

Exec-timeout 30

Logging synchronous

Do write

e.g. R1

```
R1>en
Password:
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#line cons
R1(config)#line console 0
R1(config-line)#log
R1(config-line)#login local
R1(config-line)#exec-
R1(config-line)#exec-timeout 30
R1(config-line)#loggn
R1(config-line)#logg
R1(config-line)#logging syn
R1(config-line)#logging synchronous
R1(config-line)#do write
Building configuration...
[OK]
R1(config-line)#
```

Overall completion has now risen

Completion: 23%



Dock

Check Results

2.0 VLANs and EtherChannel

This part of the lab includes the following steps:

1. In Office A, configure a Layer-2 EtherChannel named **PortChannel1** between DSW-A1 and DSW-A2 using a Cisco-proprietary protocol. Both switches should actively try to form an EtherChannel.
2. In Office B, configure a Layer-2 EtherChannel named **PortChannel1** between DSW-B1 and DSW-B2 using an open standard protocol. Both switches should actively try to form an EtherChannel.
3. Configure all links between Access and Distribution switches, including the EtherChannels, as trunk links.
 - a. Explicitly disable DTP on all ports.
 - b. Set each trunk's native VLAN to VLAN 1000 (unused).
 - c. In Office A, allow VLANs 10, 20, 40, and 99 on all trunks.
 - d. In Office B, allow VLANs 10, 20, 30, and 99 on all trunks.
4. Configure one of each office's Distribution switches as a VTPv2 server. Use domain name **JeremysITLab**.
 - a. Verify that other switches join the domain.
 - b. Configure all Access switches as VTP clients.
5. In Office A, create and name the following VLANs on one of the Distribution switches. Ensure that VTP propagates the changes.
 - a. VLAN 10: PCs

- b. VLAN 20: Phones
- c. VLAN 40: Wi-Fi
- d. VLAN 99: Management
- 6. In Office B, create and name the following VLANs on one of the Distribution switches. Ensure that VTP propagates the changes.
 - a. VLAN 10: PCs
 - b. VLAN 20: Phones
 - c. VLAN 30: Servers
 - d. VLAN 99: Management
- 7. Configure each Access switch's access port.
 - a. LWAPs will not use FlexConnect
 - b. PCs in VLAN 10, Phones in VLAN 20
 - c. SRV1 in VLAN 30
 - d. Manually configure access mode and explicitly disable DTP
- 8. Configure ASW-A1's connection to WLC1:
 - a. It must support the Wi-Fi and Management VLANs.
 - b. The Management VLAN should be untagged.
 - c. Disable DTP.
- 9. Administratively disable all unused ports on Access and Distribution switches.

2.1 Office A EtherChannel

This subsection covers the following objectives:

In Office A, configure a Layer-2 EtherChannel named **PortChannel1** between DSW-A1 and DSW-A2 using a Cisco-proprietary protocol. Both switches should actively try to form an EtherChannel.

The Cisco-proprietary protocol is PAgP (Port Aggregation Protocol) and both switches actively trying to form an EtherChannel means both should be in desirable mode.

On DSW-A1 as an example:

First need to find which interfaces are relevant, done with

Enable

Show CDP neighbors

User Access Verification

Username: cisco
Password:

DSW-A1>en

Password:

DSW-A1#sh cdp neigh

Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge

S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone

Device ID	Local Intrfce	Holdtme	Capability	Platform	Port ID
ASW-A3	Gig 1/0/3	120	S	2960	Gig 0/1
DSW-A2	Gig 1/0/5	120		3650	Gig 1/0/5
DSW-A2	Gig 1/0/4	120		3650	Gig 1/0/4
ASW-A2	Gig 1/0/2	120	S	2960	Gig 0/1
ASW-A1	Gig 1/0/1	120	S	2960	Gig 0/1
CSW1	Gig 1/1/1	120		3650	Gig 1/1/1
CSW2	Gig 1/1/2	120		3650	Gig 1/1/1

DSW-A1#

Relevant interfaces are Gig 1/0/5 and 1/0/4, so following off from checking CDP neighbours:

Configure terminal

Interface range gig 1/0/4-5

Channel-group 1 mode desirable

Do write

DSW-A1#conf t

Enter configuration commands, one per line. End with CNTL/Z.

DSW-A1(config)#int range g1/0/4-5

DSW-A1(config-if-range)#channel-group 1 mode des

DSW-A1(config-if-range)#channel-group 1 mode desirable

DSW-A1(config-if-range)#

Creating a port-channel interface Port-channel 1

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/4, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/4, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/5, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/5, changed state to up

DSW-A1(config-if-range)#do write

Building configuration...

Compressed configuration from 7383 bytes to 3601 bytes[OK]

[OK]

DSW-A1(config-if-range)#

Repeated for DSW-A2

Using do show etherchannel summary we can see that it is in use from the U flag

```
DSW-A1(config-if-range)#do sh etherchannel summary
Flags:  D - down          P - in port-channel
        I - stand-alone  s - suspended
        H - Hot-standby (LACP only)
        R - Layer3       S - Layer2
        U - in use       f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port
```

```
Number of channel-groups in use: 1
Number of aggregators:          1
```

Group	Port-channel	Protocol	Ports
1	Po1 (SU)	PAgP	Gig1/0/4 (P) Gig1/0/5 (P)

2.2 Office B EtherChannel

This subsection covers the following objectives:

In Office B, configure a Layer-2 EtherChannel named **PortChannel1** between DSW-B1 and DSW-B2 using an open standard protocol. Both switches should actively try to form an EtherChannel.

The open standard protocol is LACP (Link Aggregation Control Protocol), both should be trying to form an ether channel so both should be in active mode.

Using DSW-B1 as an example:

Recognising correct interfaces

User Access Verification

```
Username: cisco
Password:
```

```
DSW-B1>en
Password:
DSW-B1#show
DSW-B1#show cd
DSW-B1#show cdp ne
DSW-B1#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID        Local Intrfce  Holdtme    Capability  Platform  Port ID
CSW1              Gig 1/1/1      176                3650        Gig 1/1/3
CSW2              Gig 1/1/2      176                3650        Gig 1/1/3
ASW-B1            Gig 1/0/1      176                S           2960        Gig 0/1
DSW-B2            Gig 1/0/4      176                3650        Gig 1/0/4
DSW-B2            Gig 1/0/5      176                3650        Gig 1/0/5
ASW-B2            Gig 1/0/2      176                S           2960        Gig 0/1
ASW-B3            Gig 1/0/3      176                S           2960        Gig 0/1
DSW-B1#
```

Gig 1/0/4 and 1/0/5

From global configuration:

Configure terminal

Interface range gig 1/0/4-5

channel-group 1 mode active

do write

```
DSW-B1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
DSW-B1(config)#int range g1/0/4-5
DSW-B1(config-if-range)#channel-group 1 mode active
DSW-B1(config-if-range)#
Creating a port-channel interface Port-channel 1

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/4, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/4, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/5, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/5, changed state to up

DSW-B1(config-if-range)#do wr
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
```

Repeated for DSW-B2

Can see from EtherChannel summary that it is in use from the U flag

```
DSW-B2(config-if-range)#do sh etherchannel summary
Flags:  D - down          P - in port-channel
        I - stand-alone  s - suspended
        H - Hot-standby (LACP only)
        R - Layer3       S - Layer2
        U - in use       f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port
```

Number of channel-groups in use: 1

Number of aggregators: 1

Group	Port-channel	Protocol	Ports
1	Po1(SU)	LACP	Gig1/0/4(P) Gig1/0/5(P)

```
DSW-B2(config-if-range)#
```

2.3 Trunk Links

This subsection covers the following objectives:

Configure all links between Access and Distribution switches, including the EtherChannels, as trunk links.

- a. Explicitly disable DTP on all ports.
- b. Set each trunk's native VLAN to VLAN 1000 (unused).
- c. In Office A, allow VLANs 10, 20, 40, and 99 on all trunks.
- d. In Office B, allow VLANs 10, 20, 30, and 99 on all trunks.

2.3.1 Office A

Using DSW-A1 as an example:

First we find the relevant interfaces via

Show cdp neighbors

```
DSW-A1#sh cdp nei
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID      Local Intrfce  Holdtme    Capability  Platform  Port ID
ASW-A2         Gig 1/0/2        123        S           2960      Gig 0/1
ASW-A1         Gig 1/0/1        123        S           2960      Gig 0/1
CSW1           Gig 1/1/1        123                3650      Gig 1/1/1
ASW-A3         Gig 1/0/3        123        S           2960      Gig 0/1
DSW-A2         Por 1            123                3650      Gig 1/0/4
DSW-A2         Por 1            123                3650      Gig 1/0/5
DSW-A2         Por 1            123                3650      Por 1
CSW2           Gig 1/1/2        123                3650      Gig 1/1/1
DSW-A1#
```

Gig 1/0/1-3 are the relevant interfaces alongside the portchannels

Interface range gig 1/0/1-3

Switchport mode trunk

Switchport nonegotiate

Switchport trunk native vlan 1000

Switchport trunk allowed vlan 10,20,40,99

```
DSW-A1(config-if-range)#Switchport nonegotiate
DSW-A1(config-if-range)#Switchport trunk native vlan 1000
DSW-A1(config-if-range)#Switchport trunk allowed vlan 10,20,40,99
DSW-A1(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/2, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/3, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/3, changed state to up
DSW-A1(config-if-range)#do wr
```

Repeated for the etherchannel, accessed using

Interface port 1

This is then repeated with the access switches

Using ASW-A1 as an example

Show cdp neighbors

```

ASW-A1#sh cdp n
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID      Local Intrfce  Holdtme    Capability  Platform  Port ID
DSW-A1         Gig 0/1        147        S           3650      Gig 1/0/1
DSW-A2         Gig 0/2        172        S           3650      Gig 1/0/1

```

Relevant interfaces are g0/1-2

Interface range g0/1-2

Switchport mode trunk

Switchport nonegotiate

Switchport trunk native vlan 1000

Switchport trunk allowed vlan 10,20,40,99

do write

```

ASW-A1(config)#Interface range g0/1-2
ASW-A1(config-if-range)#Switchport mode trunk

```

```

ASW-A1(config-if-range)#Switchport nonegotiate
ASW-A1(config-if-range)#Switchport trunk native vlan 1000
ASW-A1(config-if-range)#Switchport trunk allowed vlan 10,20,40,99
ASW-A1(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
do wr
Building configuration...
[OK]

```

2.3.2 Office B

Office B is the exact same with the exception of allowing VLAN 30 instead of 40

Using DSW-B1 as an example:

Show cdp neighbors

```

DSW-B1#sh cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID      Local Intrfce  Holdtme    Capability  Platform  Port ID
ASW-B1         Gig 1/0/1        129        S           2960      Gig 0/1
CSW1           Gig 1/1/1        129        S           3650      Gig 1/1/3
DSW-B2         Por 1            129        S           3650      Gig 1/0/4
DSW-B2         Por 1            129        S           3650      Gig 1/0/5
DSW-B2         Por 1            129        S           3650      Por 1
ASW-B2         Gig 1/0/2        129        S           2960      Gig 0/1
ASW-B3         Gig 1/0/3        129        S           2960      Gig 0/1
CSW2           Gig 1/1/2        129        S           3650      Gig 1/1/3

```

Relevant Interfaces are Gig 1/0/1-3

Interface range gig 1/0/1-3

Switchport mode trunk

Switchport nonegotiate

Switchport trunk native vlan 1000

Switchport trunk allowed vlan 10,20,30,99

```
DSW-B1(config)#Interface range gig 1/0/1-3
DSW-B1(config-if-range)#Switchport mode trunk

DSW-B1(config-if-range)#Switchport nonegotiate
DSW-B1(config-if-range)#Switchport trunk native vlan 1000
DSW-B1(config-if-range)#Switchport trunk allowed vlan 10,20,30,99
DSW-B1(config-if-range)#
```

Repeated for interface por 1

```
DSW-B1(config)#int pl
DSW-B1(config-if)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet1/0/2 (1000),
with ASW-B2 GigabitEthernet0/1 (1).

DSW-B1(config-if)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet1/0/1 (1000),
with ASW-B1 GigabitEthernet0/1 (1).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet1/0/3 (1000),
with ASW-B3 GigabitEthernet0/1 (1).

DSW-B1(config-if)#Switchport mode trunk

DSW-B1(config-if)#Switchport nonegotiate
DSW-B1(config-if)#Switchport trunk native vlan 1000
DSW-B1(config-if)#Switchport trunk allowed vlan 10,20,30,99
DSW-B1(config-if)#do wr
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
```

Need to then repeat for the access switches. Using ASW-B1 as an example:

Show cdp neighbors

```
ASW-B1#sh cdp n
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID        Local Intrfce    Holdtme    Capability    Platform    Port ID
DSW-B1           Gig 0/1             121        R, S, T, B    3650        Gig 1/0/1
DSW-B2           Gig 0/2             129        R, S, T, B    3650        Gig 1/0/1
```

Relevant ports are g0/1-2

Interface range g0/1-2

Switchport mode trunk

Switchport nonegotiate

Switchport trunk native vlan 1000

Switchport trunk allowed vlan 10,20,30,99

do write

```

ASW-B1(config)#Interface range g0/1-2
ASW-B1(config-if-range)#Switchport mode trunk

ASW-B1(config-if-range)#Switchport nonegotiate
ASW-B1(config-if-range)#Switchport trunk native vlan 1000
ASW-B1(config-if-range)#Switchport trunk allowed vlan 10,20,30,99
ASW-B1(config-if-range)#do write
Building configuration...
[OK]
ASW-B1(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

```

2.4 VTPv2 Servers

The objectives of this subsection are:

Configure one of each office's Distribution switches as a VTPv2 server. Use domain name **JeremysITLab**.

- Verify that other switches join the domain.
- Configure all Access switches as VTP clients.

On DSW-A1:

Enable

Show vtp status (this shows us the default configuration)

```

DSW-A1#sh vtp status
VTP Version capable      : 1 to 2
VTP version running      : 1
VTP Domain Name          :
VTP Pruning Mode         : Disabled
VTP Traps Generation     : Disabled
Device ID                 : 0001.6395.E700
Configuration last modified by 0.0.0.0 at 0-0-00 00:00:00
Local updater ID is 0.0.0.0 (no valid interface found)

Feature VLAN :
-----
VTP Operating Mode       : Server
Maximum VLANs supported locally : 1005
Number of existing VLANs : 5
Configuration Revision   : 0
MD5 digest               : 0x7D 0x5A 0xA6 0x0E 0x9A 0x72 0xA0 0x3A
                        : 0xF0 0x58 0x10 0x6C 0x9C 0x0F 0xA0 0xF7

```

Configure terminal

Vtp domain JeremysITLab

Vtp version 2

Do write


```

DSW-A1(config)#vtp domain JeremysITLab
Changing VTP domain name from NULL to JeremysITLab
DSW-A1(config)#vtp version 2
DSW-A1(config)#do write
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]

```

Checking DSW-A2, we can see the changes have propagated

```

DSW-A2#sh vtp stat
VTP Version capable          : 1 to 2
VTP version running         : 2
VTP Domain Name              : JeremysITLab
VTP Pruning Mode             : Disabled
VTP Traps Generation         : Disabled
Device ID                    : 0001.6492.C600
Configuration last modified by 0.0.0.0 at 2-28-93 15:11:55
Local updater ID is 0.0.0.0 (no valid interface found)

```

The process is the exact same for office B

Configuration done on DSW-B1

```

DSW-B1(config)#Vtp domain JeremysITLab
Changing VTP domain name from NULL to JeremysITLab
DSW-B1(config)#Vtp version 2
DSW-B1(config)#Do write

```

Verifying propagation on DSW-B2

```

DSW-B2#sh vtp status
VTP Version capable          : 1 to 2
VTP version running         : 2
VTP Domain Name              : JeremysITLab
VTP Pruning Mode             : Disabled
VTP Traps Generation         : Disabled
Device ID                    : 0007.EC8C.4D00
Configuration last modified by 0.0.0.0 at 2-28-93 15:19:42
Local updater ID is 0.0.0.0 (no valid interface found)

```

To make all the access switches VTP clients the following command in configure terminal is used:

Vtp mode client

Before

```
ASW-A1(config)#do sh vtp stat
VTP Version capable      : 1 to 2
VTP version running      : 2
VTP Domain Name          : JeremysITLab
VTP Pruning Mode         : Disabled
VTP Traps Generation     : Disabled
Device ID                : 0002.4AB5.E200
Configuration last modified by 0.0.0.0 at 2-28-93 15:11:55
Local updater ID is 0.0.0.0 (no valid interface found)
```

Feature VLAN :

VTP Operating Mode : **Server**

After

```
ASW-A1(config)#vtp mode client
Setting device to VTP CLIENT mode.
ASW-A1(config)#do sh vtp status
VTP Version capable      : 1 to 2
VTP version running      : 2
VTP Domain Name          : JeremysITLab
VTP Pruning Mode         : Disabled
VTP Traps Generation     : Disabled
Device ID                : 0002.4AB5.E200
Configuration last modified by 0.0.0.0 at 2-28-93 15:11:55
```

Feature VLAN :

VTP Operating Mode : **Client**

Repeated for every access switch

2.5 VLAN Configuration/Propagation Via VTP

This subsection includes the following objectives:

In Office A, create and name the following VLANs on one of the Distribution switches. Ensure that VTP propagates the changes.

- a. VLAN 10: PCs
- b. VLAN 20: Phones
- c. VLAN 40: Wi-Fi
- d. VLAN 99: Management

In Office B, create and name the following VLANs on one of the Distribution switches. Ensure that VTP propagates the changes.

- a. VLAN 10: PCs
- b. VLAN 20: Phones
- c. VLAN 30: Servers
- d. VLAN 99: Management

2.5.1 Office A

For each VLAN after configure terminal:

Vlan <VLAN NUMBER>

Name <VLAN NAME>

DSW-A1:

```
DSW-A1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
DSW-A1(config)#vlan 10
DSW-A1(config-vlan)#name PCs
DSW-A1(config-vlan)#ex
DSW-A1(config)#vlan 20
DSW-A1(config-vlan)#name Phones
DSW-A1(config-vlan)#ex
DSW-A1(config)#vlan 40
DSW-A1(config-vlan)#name Wi-Fi
DSW-A1(config-vlan)#ex
DSW-A1(config)#vlan 99
DSW-A1(config-vlan)#Management
^
% Invalid input detected at '^' marker.

DSW-A1(config-vlan)#name Management
DSW-A1(config-vlan)#do write
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
```

Checking DSW-A2 we can see the changes have propagated

DSW-A2#sh vlan brief

VLAN	Name	Status	Ports
1	default	active	Gig1/0/6, Gig1/0/7, Gig1/0/8, Gig1/0/9 Gig1/0/10, Gig1/0/11, Gig1/0/12, Gig1/0/13 Gig1/0/14, Gig1/0/15, Gig1/0/16, Gig1/0/17 Gig1/0/18, Gig1/0/19, Gig1/0/20, Gig1/0/21 Gig1/0/22, Gig1/0/23, Gig1/0/24, Gig1/1/1 Gig1/1/2, Gig1/1/3, Gig1/1/4
10	PCs	active	
20	Phones	active	
40	Wi-Fi	active	
99	Management	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

2.5.2 Office B

Exact same as Office A except instead of VLAN 40: Wi-Fi we have VLAN 30: Servers

DSW-B1

```

DSW-B1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
DSW-B1(config)#vlan 10
DSW-B1(config-vlan)#name PCs
DSW-B1(config-vlan)#ex
DSW-B1(config)#vlan 20
DSW-B1(config-vlan)#name Phones
DSW-B1(config-vlan)#ex
DSW-B1(config)#vlan 30
DSW-B1(config-vlan)#name Servers
DSW-B1(config-vlan)#ex
DSW-B1(config)#vlan 99
DSW-B1(config-vlan)#name Management
DSW-B1(config-vlan)#do write
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]

```

Propagated changes on DSW-B2:

```
DSW-B2#sh vlan br
```

VLAN	Name	Status	Ports
1	default	active	Gig1/0/6, Gig1/0/7, Gig1/0/8, Gig1/0/9 Gig1/0/10, Gig1/0/11, Gig1/0/12, Gig1/0/13 Gig1/0/14, Gig1/0/15, Gig1/0/16, Gig1/0/17 Gig1/0/18, Gig1/0/19, Gig1/0/20, Gig1/0/21 Gig1/0/22, Gig1/0/23, Gig1/0/24, Gig1/1/1 Gig1/1/2, Gig1/1/3, Gig1/1/4
10	PCs	active	
20	Phones	active	
30	Servers	active	
99	Management	active	

2.6 Access Port Configuration

This subsection covers the following objectives:

Configure each Access switch's access port.

- a. Manually configure access mode and explicitly disable DTP
- b. LWAPs will not use FlexConnect
- c. PCs in VLAN 10, Phones in VLAN 20
- d. SRV1 in VLAN 30

part a applies to every switches access port, it is done via:

```

enable
configure terminal
interface <RELEVANT INTERFACE>
switchport mode access
switchport nonegotiate

```

parts b, c and d are specific to switches connected to specific endpoints

2.6.1 LWAPs

The relevant switches here are ASW-A1 and ASW-B1 in this configuration



The LWAP doesn't use CDP so we can't use the `show CDP neighbours` command to find the correct interface, instead, we take note of its mac address and then use the command

Show mac-address-table

To see which is the relevant interface

Using ASW-A1 as an example:

LWAP mac address

INTERFACE	Bandwidth	Duplex	MAC Address
GigabitEthernet0	1000 Mbps	Full Duplex	0002.4AE7.1601
Dot11Radio0	100 Mbps	Full Duplex	

Show mac-address-table

```
ASW-A1#sh mac-address-table
```

Mac Address Table			
Vlan	Mac Address	Type	Ports
1	0002.4ae7.1601	DYNAMIC	Fa0/1
10	0001.6395.e701	DYNAMIC	Gig0/1
20	0001.6395.e701	DYNAMIC	Gig0/1
40	0001.6395.e701	DYNAMIC	Gig0/1
99	0001.6395.e701	DYNAMIC	Gig0/1

Relevant interface is Fa0/1

Now we have the correct interface we can proceed

Configure terminal

Interface f0/1

Switchport mode access

Switchport nonegotiate

Switchport access vlan 99

```
ASW-A1#Configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
ASW-A1(config)#Interface f0/1
ASW-A1(config-if)#Switchport mode access
ASW-A1(config-if)#Switchport nonegotiate
ASW-A1(config-if)#Switchport access vlan 99
ASW-A1(config-if)#do write
Building configuration...
[OK]
```

Port	Name	Status	Vlan
Fa0/1		connected	99

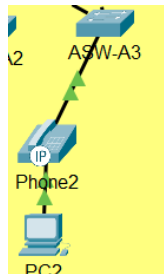
ASW-B1:

```
ASW-B1#Configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
ASW-B1(config)#Interface f0/1
ASW-B1(config-if)#Switchport mode access
ASW-B1(config-if)#Switchport nonegotiate
ASW-B1(config-if)#Switchport access vlan 99
ASW-B1(config-if)#do write
Building configuration...
[OK]
```

99	Management	active	Fa0/1
----	------------	--------	-------

2.6.2 PCs/Phones

Relevant switches are ASW-A2, ASW-A3 and ASW-B2 in the following configuration



Using ASW-A2 as an example

We first find the relevant interface

Show cdp neighbors gives this output

```
ASW-A2#sh cdp neigh
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID        Local Intrfce  Holdtme  Capability  Platform  Port ID
DSW-A2           Gig 0/2        130      S           3650      Gig 1/0/2
DSW-A1           Gig 0/1        130      S           3650      Gig 1/0/2
Switch           Fas 0/1        129      H P         7960
```

We can see 3 connections, 2 are to the distribution switches and 1 is titled switch

Using show interface status, we can see there are 3 interfaces in use

```

ASW-A2#sh int status
Port      Name      Status      Vlan      Duplex  Speed  Type
Fa0/1     Fa0/1     connected   1         auto    auto   10/100BaseTX
Fa0/2     Fa0/2     notconnect  1         auto    auto   10/100BaseTX
Fa0/3     Fa0/3     notconnect  1         auto    auto   10/100BaseTX
Fa0/4     Fa0/4     notconnect  1         auto    auto   10/100BaseTX
Fa0/5     Fa0/5     notconnect  1         auto    auto   10/100BaseTX
Fa0/6     Fa0/6     notconnect  1         auto    auto   10/100BaseTX
Fa0/7     Fa0/7     notconnect  1         auto    auto   10/100BaseTX
Fa0/8     Fa0/8     notconnect  1         auto    auto   10/100BaseTX
Fa0/9     Fa0/9     notconnect  1         auto    auto   10/100BaseTX
Fa0/10    Fa0/10    notconnect  1         auto    auto   10/100BaseTX
Fa0/11    Fa0/11    notconnect  1         auto    auto   10/100BaseTX
Fa0/12    Fa0/12    notconnect  1         auto    auto   10/100BaseTX
Fa0/13    Fa0/13    notconnect  1         auto    auto   10/100BaseTX
Fa0/14    Fa0/14    notconnect  1         auto    auto   10/100BaseTX
Fa0/15    Fa0/15    notconnect  1         auto    auto   10/100BaseTX
Fa0/16    Fa0/16    notconnect  1         auto    auto   10/100BaseTX
Fa0/17    Fa0/17    notconnect  1         auto    auto   10/100BaseTX
Fa0/18    Fa0/18    notconnect  1         auto    auto   10/100BaseTX
Fa0/19    Fa0/19    notconnect  1         auto    auto   10/100BaseTX
Fa0/20    Fa0/20    notconnect  1         auto    auto   10/100BaseTX
Fa0/21    Fa0/21    notconnect  1         auto    auto   10/100BaseTX
Fa0/22    Fa0/22    notconnect  1         auto    auto   10/100BaseTX
Fa0/23    Fa0/23    notconnect  1         auto    auto   10/100BaseTX
Fa0/24    Fa0/24    notconnect  1         auto    auto   10/100BaseTX
Gig0/1    Gig0/1    connected   trunk     auto    auto   10/100BaseTX
Gig0/2    Gig0/2    connected   trunk     auto    auto   10/100BaseTX

```

If 2 are being used for connections to the distribution switch connections (the gig interfaces) then the remaining Fa0/1 must be the one connected to the IP phone, continuing on:

Configure terminal

Interface f0/1

Switchport mode access

Switchport nonegotiate

Switchport access vlan 10

Switchport voice vlan 20

Do write

```

ASW-A2#Configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
ASW-A2 (config)#Interface f0/1
ASW-A2 (config-if)#Switchport mode access
ASW-A2 (config-if)#Switchport nonegotiate
ASW-A2 (config-if)#Switchport access vlan 10
ASW-A2 (config-if)#Switchport voice vlan 20
ASW-A2 (config-if)#Do write
Building configuration...
[OK]

```

Repeated exactly for A3 and B2 as they also use fa0/1 as their relevant interface

2.6.3 SRV1

The relevant switch here is ASW-B3



We can easily deduce the relevant interface using the previous approach as from 2.6.2

```
ASW-B3#sh int stat
Port      Name      Status      Vlan      Duplex  Speed  Type
Fa0/1     Name      connected   1         auto    auto   10/100BaseTX
Fa0/2     notconnect 1         auto    auto   10/100BaseTX
Fa0/3     notconnect 1         auto    auto   10/100BaseTX
Fa0/4     notconnect 1         auto    auto   10/100BaseTX
Fa0/5     notconnect 1         auto    auto   10/100BaseTX
Fa0/6     notconnect 1         auto    auto   10/100BaseTX
Fa0/7     notconnect 1         auto    auto   10/100BaseTX
Fa0/8     notconnect 1         auto    auto   10/100BaseTX
Fa0/9     notconnect 1         auto    auto   10/100BaseTX
Fa0/10    notconnect 1         auto    auto   10/100BaseTX
Fa0/11    notconnect 1         auto    auto   10/100BaseTX
Fa0/12    notconnect 1         auto    auto   10/100BaseTX
Fa0/13    notconnect 1         auto    auto   10/100BaseTX
Fa0/14    notconnect 1         auto    auto   10/100BaseTX
Fa0/15    notconnect 1         auto    auto   10/100BaseTX
Fa0/16    notconnect 1         auto    auto   10/100BaseTX
Fa0/17    notconnect 1         auto    auto   10/100BaseTX
Fa0/18    notconnect 1         auto    auto   10/100BaseTX
Fa0/19    notconnect 1         auto    auto   10/100BaseTX
Fa0/20    notconnect 1         auto    auto   10/100BaseTX
Fa0/21    notconnect 1         auto    auto   10/100BaseTX
Fa0/22    notconnect 1         auto    auto   10/100BaseTX
Fa0/23    notconnect 1         auto    auto   10/100BaseTX
Fa0/24    notconnect 1         auto    auto   10/100BaseTX
Gig0/1    connected  trunk     auto    auto   10/100BaseTX
Gig0/2    connected  trunk     auto    auto   10/100BaseTX
```

```
ASW-B3#sh cdp neigh
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID      Local Intrfce  Holdtme  Capability  Platform  Port ID
DSW-B1         Gig 0/1       129      R           3650      Gig 1/0/3
DSW-B2         Gig 0/2       130      R           3650      Gig 1/0/3
```

Fa0/1 is the relevant interface, from here:

Configure terminal

Interface f0/1

Switchport mode access

switchport nonegotiate

switchport access vlan 30

do write

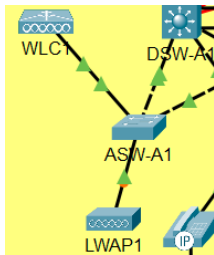
2.7 WLC1

This subsection covers the following objectives:

Configure ASW-A1's connection to WLC1:

- a. It must support the Wi-Fi and Management VLANs.
- b. The Management VLAN should be untagged.
- c. Disable DTP.

This switch was previously configured as it has an LWAP connected



Using show interface status, we get this output

```
ASW-A1#sh int status
Port      Name      Status      Vlan      Duplex  Speed  Type
Fa0/1     Fa0/1     connected   99        auto    auto   10/100BaseTX
Fa0/2     Fa0/2     connected   1         auto    auto   10/100BaseTX
Fa0/3     Fa0/3     notconnect  1         auto    auto   10/100BaseTX
Fa0/4     Fa0/4     notconnect  1         auto    auto   10/100BaseTX
Fa0/5     Fa0/5     notconnect  1         auto    auto   10/100BaseTX
Fa0/6     Fa0/6     notconnect  1         auto    auto   10/100BaseTX
Fa0/7     Fa0/7     notconnect  1         auto    auto   10/100BaseTX
Fa0/8     Fa0/8     notconnect  1         auto    auto   10/100BaseTX
Fa0/9     Fa0/9     notconnect  1         auto    auto   10/100BaseTX
Fa0/10    Fa0/10    notconnect  1         auto    auto   10/100BaseTX
Fa0/11    Fa0/11    notconnect  1         auto    auto   10/100BaseTX
Fa0/12    Fa0/12    notconnect  1         auto    auto   10/100BaseTX
Fa0/13    Fa0/13    notconnect  1         auto    auto   10/100BaseTX
Fa0/14    Fa0/14    notconnect  1         auto    auto   10/100BaseTX
Fa0/15    Fa0/15    notconnect  1         auto    auto   10/100BaseTX
Fa0/16    Fa0/16    notconnect  1         auto    auto   10/100BaseTX
Fa0/17    Fa0/17    notconnect  1         auto    auto   10/100BaseTX
Fa0/18    Fa0/18    notconnect  1         auto    auto   10/100BaseTX
Fa0/19    Fa0/19    notconnect  1         auto    auto   10/100BaseTX
Fa0/20    Fa0/20    notconnect  1         auto    auto   10/100BaseTX
Fa0/21    Fa0/21    notconnect  1         auto    auto   10/100BaseTX
Fa0/22    Fa0/22    notconnect  1         auto    auto   10/100BaseTX
Fa0/23    Fa0/23    notconnect  1         auto    auto   10/100BaseTX
Fa0/24    Fa0/24    notconnect  1         auto    auto   10/100BaseTX
Gig0/1    Gig0/1    connected   trunk     auto    auto   10/100BaseTX
Gig0/2    Gig0/2    connected   trunk     auto    auto   10/100BaseTX
```

The 2 Gig interfaces are for the distribution switch connection, Fa0/1 is the LWAP connection leaving Fa0/2 as the WLC connection. From here:

Configure terminal

Interface f0/2

Switchport mode trunk

Switchport nonegotiate

Switchport trunk allowed vlan 40,99

Switchport trunk native vlan 99

Do write

```
ASW-A1(config-if)#Switchport mode trunk
ASW-A1(config-if)#Switchport nonegotiate
ASW-A1(config-if)#Switchport trunk allowed vlan 40,99
ASW-A1(config-if)#Switchport trunk native vlan 99
ASW-A1(config-if)#Do write
Building configuration...
[OK]
```

2.8 Disabling Unused Ports

The following subsection covers this topic:

Administratively disable all unused ports on Access and Distribution switches

Using DSW-A1 as an example:

We can check which ports aren't in use with

Show interface status

```
DSW-A1#sh int status
```

Port	Name	Status	Vlan	Duplex	Speed	Type
Pol		connected	trunk	auto	auto	
Gig1/0/1		connected	trunk	auto	auto	10/100BaseTX
Gig1/0/2		connected	trunk	auto	auto	10/100BaseTX
Gig1/0/3		connected	trunk	auto	auto	10/100BaseTX
Gig1/0/4		connected	trunk	auto	auto	10/100BaseTX
Gig1/0/5		connected	trunk	auto	auto	10/100BaseTX
Gig1/0/6		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/7		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/8		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/9		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/10		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/11		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/12		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/13		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/14		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/15		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/16		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/17		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/18		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/19		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/20		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/21		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/22		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/23		notconnect	1	auto	auto	10/100BaseTX
Gig1/0/24		notconnect	1	auto	auto	10/100BaseTX
Gig1/1/1		connected	1	auto	auto	10/100BaseTX
Gig1/1/2		connected	1	auto	auto	10/100BaseTX
Gig1/1/3		notconnect	1	auto	auto	10/100BaseTX
Gig1/1/4		notconnect	1	auto	auto	10/100BaseTX

Every port with the notconnect status needs to be disabled

Done with:

Configure terminal

Interface range g1/0/6-24,g1/1/3-4

Shutdown

Do write

Interface status output shows it was successful

```
DSW-A1(config-if-range)#do sh int status
Port      Name      Status      Vlan      Duplex  Speed  Type
Po1                connected   trunk      auto     auto
Gig1/0/1          connected   trunk      auto     auto    10/100BaseTX
Gig1/0/2          connected   trunk      auto     auto    10/100BaseTX
Gig1/0/3          connected   trunk      auto     auto    10/100BaseTX
Gig1/0/4          connected   trunk      auto     auto    10/100BaseTX
Gig1/0/5          connected   trunk      auto     auto    10/100BaseTX
Gig1/0/6          disabled 1          auto     auto    10/100BaseTX
Gig1/0/7          disabled 1          auto     auto    10/100BaseTX
Gig1/0/8          disabled 1          auto     auto    10/100BaseTX
Gig1/0/9          disabled 1          auto     auto    10/100BaseTX
Gig1/0/10         disabled 1          auto     auto    10/100BaseTX
Gig1/0/11         disabled 1          auto     auto    10/100BaseTX
Gig1/0/12         disabled 1          auto     auto    10/100BaseTX
Gig1/0/13         disabled 1          auto     auto    10/100BaseTX
Gig1/0/14         disabled 1          auto     auto    10/100BaseTX
Gig1/0/15         disabled 1          auto     auto    10/100BaseTX
Gig1/0/16         disabled 1          auto     auto    10/100BaseTX
Gig1/0/17         disabled 1          auto     auto    10/100BaseTX
Gig1/0/18         disabled 1          auto     auto    10/100BaseTX
Gig1/0/19         disabled 1          auto     auto    10/100BaseTX
Gig1/0/20         disabled 1          auto     auto    10/100BaseTX
Gig1/0/21         disabled 1          auto     auto    10/100BaseTX
Gig1/0/22         disabled 1          auto     auto    10/100BaseTX
Gig1/0/23         disabled 1          auto     auto    10/100BaseTX
Gig1/0/24         disabled 1          auto     auto    10/100BaseTX
Gig1/1/1          connected 1          auto     auto    10/100BaseTX
Gig1/1/2          connected 1          auto     auto    10/100BaseTX
Gig1/1/3          disabled 1          auto     auto    10/100BaseTX
Gig1/1/4          disabled 1          auto     auto    10/100BaseTX
```

Process repeated with every switch

Example using an access switch (ASW-A1):

Port	Name	Status	Vlan	Duplex	Speed	Type
Fa0/1		connected	99	auto	auto	10/100BaseTX
Fa0/2		connected	trunk	auto	auto	10/100BaseTX
Fa0/3		notconnect	1	auto	auto	10/100BaseTX
Fa0/4		notconnect	1	auto	auto	10/100BaseTX
Fa0/5		notconnect	1	auto	auto	10/100BaseTX
Fa0/6		notconnect	1	auto	auto	10/100BaseTX
Fa0/7		notconnect	1	auto	auto	10/100BaseTX
Fa0/8		notconnect	1	auto	auto	10/100BaseTX
Fa0/9		notconnect	1	auto	auto	10/100BaseTX
Fa0/10		notconnect	1	auto	auto	10/100BaseTX
Fa0/11		notconnect	1	auto	auto	10/100BaseTX
Fa0/12		notconnect	1	auto	auto	10/100BaseTX
Fa0/13		notconnect	1	auto	auto	10/100BaseTX
Fa0/14		notconnect	1	auto	auto	10/100BaseTX
Fa0/15		notconnect	1	auto	auto	10/100BaseTX
Fa0/16		notconnect	1	auto	auto	10/100BaseTX
Fa0/17		notconnect	1	auto	auto	10/100BaseTX
Fa0/18		notconnect	1	auto	auto	10/100BaseTX
Fa0/19		notconnect	1	auto	auto	10/100BaseTX
Fa0/20		notconnect	1	auto	auto	10/100BaseTX
Fa0/21		notconnect	1	auto	auto	10/100BaseTX
Fa0/22		notconnect	1	auto	auto	10/100BaseTX
Fa0/23		notconnect	1	auto	auto	10/100BaseTX
Fa0/24		notconnect	1	auto	auto	10/100BaseTX
Gig0/1		connected	trunk	auto	auto	10/100BaseTX
Gig0/2		connected	trunk	auto	auto	10/100BaseTX

Configure terminal

Interface range f0/3-24

Shutdown

Do write

Port	Name	Status	Vlan	Duplex	Speed	Type
Fa0/1		connected	99	auto	auto	10/100BaseTX
Fa0/2		connected	trunk	auto	auto	10/100BaseTX
Fa0/3		disabled	1	auto	auto	10/100BaseTX
Fa0/4		disabled	1	auto	auto	10/100BaseTX
Fa0/5		disabled	1	auto	auto	10/100BaseTX
Fa0/6		disabled	1	auto	auto	10/100BaseTX
Fa0/7		disabled	1	auto	auto	10/100BaseTX
Fa0/8		disabled	1	auto	auto	10/100BaseTX
Fa0/9		disabled	1	auto	auto	10/100BaseTX
Fa0/10		disabled	1	auto	auto	10/100BaseTX
Fa0/11		disabled	1	auto	auto	10/100BaseTX
Fa0/12		disabled	1	auto	auto	10/100BaseTX
Fa0/13		disabled	1	auto	auto	10/100BaseTX
Fa0/14		disabled	1	auto	auto	10/100BaseTX
Fa0/15		disabled	1	auto	auto	10/100BaseTX
Fa0/16		disabled	1	auto	auto	10/100BaseTX
Fa0/17		disabled	1	auto	auto	10/100BaseTX
Fa0/18		disabled	1	auto	auto	10/100BaseTX
Fa0/19		disabled	1	auto	auto	10/100BaseTX
Fa0/20		disabled	1	auto	auto	10/100BaseTX
Fa0/21		disabled	1	auto	auto	10/100BaseTX
Fa0/22		disabled	1	auto	auto	10/100BaseTX
Fa0/23		disabled	1	auto	auto	10/100BaseTX
Fa0/24		disabled	1	auto	auto	10/100BaseTX
Gig0/1		connected	trunk	auto	auto	10/100BaseTX
Gig0/2		connected	trunk	auto	auto	10/100BaseTX

ASW-A1 has 4 connection vs every other access switches 3 so the interface range is different, including 1 extra interface

Configure terminal
Interface range f0/2-24
Shutdown
Do write

That is Part Layer 2 complete, completion change:

Completion: 32%
☒ Dock [Check Results](#)

Completion: 50%
☒ Dock [Check Results](#)

3.0 Layer 3